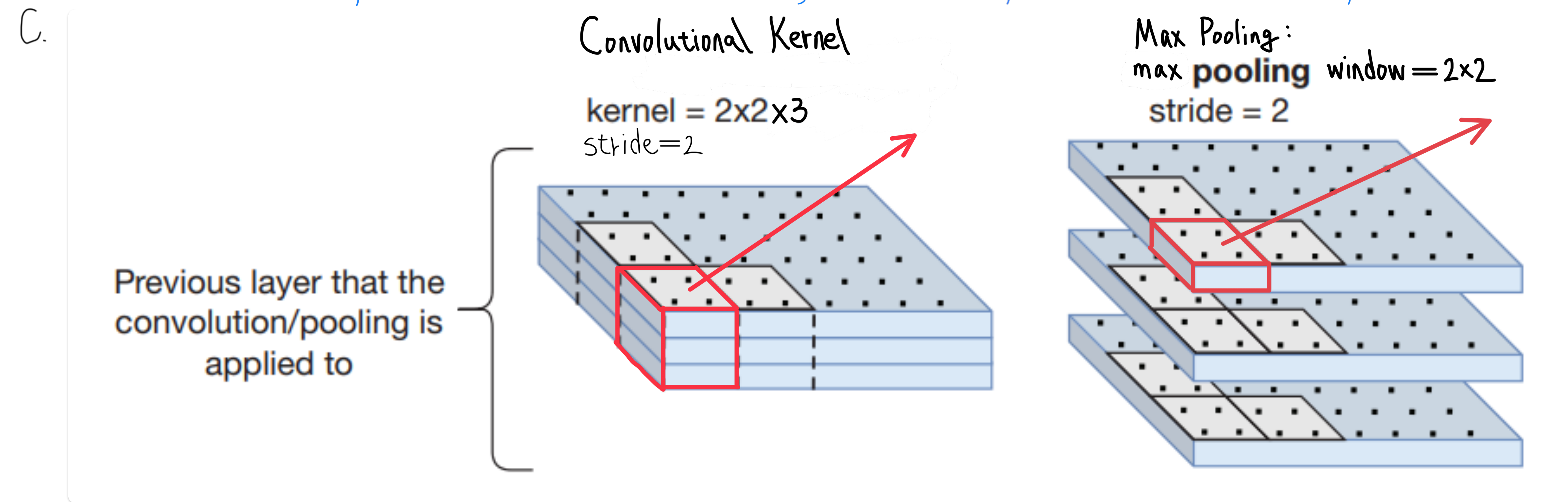


1. Convolutional Kernel Operation vs. Max Pooling Window Operation:

- A. a convolutional kernel in the current convolutional layer combines the outputs from sets of neurons across the multiple output channels of the preceding convolutional layer as shown on the left of figure below, a $2 \times 2 \times 3$ kernel with a stride of 2 bundles together all 3 output channels of the preceding convolutional layer to compute the inputs for a single neuron in the current convolutional layer. consequently, the num of output channels in the current convolutional layer is decoupled from the num of output channels in the preceding convolutional layer
- B. a max pooling window extracts outputs from a set of neurons strictly with a single output channel of the preceding convolutional layer as shown on the right of figure below, a 2×2 max pooling window with a stride of 2 independently perform maximum extraction with one output channel of the preceding convolutional layer to compute the inputs for a single neuron in the current convolutional layer. therefore, the num of output channels in the current convolutional layer remains strictly identical to the num of output channels in the preceding convolutional layer



2. A Role of Max Pooling:

max pooling is a way to reduce the spatial size of a convolutional layer by decreasing the size of feature maps in that convolutional layer. it can be used as an alternative to a large stride.